



Week 6 Lesson Plan

Broadcasts and Game Over

Big idea: Sprites can't see each other's code, but they can shout a message that everyone hears. That's how the different parts of a program work together.

At a glance

New concepts	Message passing, program state, coordinating multiple sprites
New blocks	<code>broadcast</code> , <code>broadcast and wait</code> , <code>when I receive</code> , <code>switch backdrop to</code> , <code>stop all</code> , <code>stop this script</code> , <code>show</code> , <code>hide</code>
Class activity	Magic Wand, then Two-Scene Story
Project checkpoint	Lives, a title screen, a game over screen, and a restart
Homework	A win condition; a restart

Before the class

- Print the Week 6 handout.
- Skeleton `Sky Catcher – end of Week 5` ready.
- This class has the most moving parts of any in the course. Read it twice.

0:00–0:10 – Show & Tell

Ask for high scores specifically. Get two or three up on the projector and let the class try to beat them. Five minutes of competitive play is worth it — it's the clearest signal all term that they've built something real.

0:10–0:25 – Live demo: broadcasts and backdrops

Build live with two sprites:

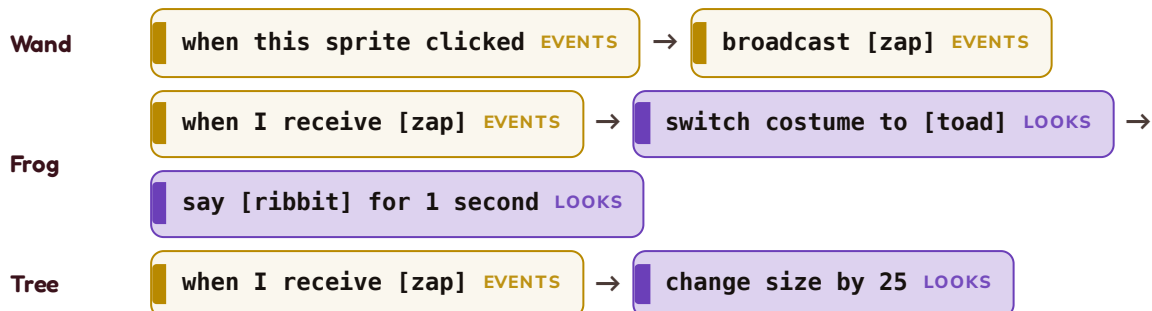
1. On sprite A: `when this sprite clicked` → `broadcast [transform]`.
2. On sprite B: `when I receive [transform]` → `switch costume to [something]` → `change size by 50`.
3. Click sprite A. Sprite B reacts. Emphasise that sprite A has no idea sprite B exists.
4. Add a **third** sprite that also listens for `transform` and does something different. One shout, three reactions.
5. Show `switch backdrop to [Game Over]` inside a `when I receive`.
6. Show `stop all` and what it does. Then `stop this script` and the difference.
7. **Deliberate mistake:** broadcast a message that nothing is listening for (typo in the message name, or two similarly-named messages). Nothing happens. Be confused, then find it in the dropdown.

That typo bug is extremely common today because message names are typed by hand. Warn them.

0:25–0:55 – Class activity

Activity A – Magic Wand (14 min)

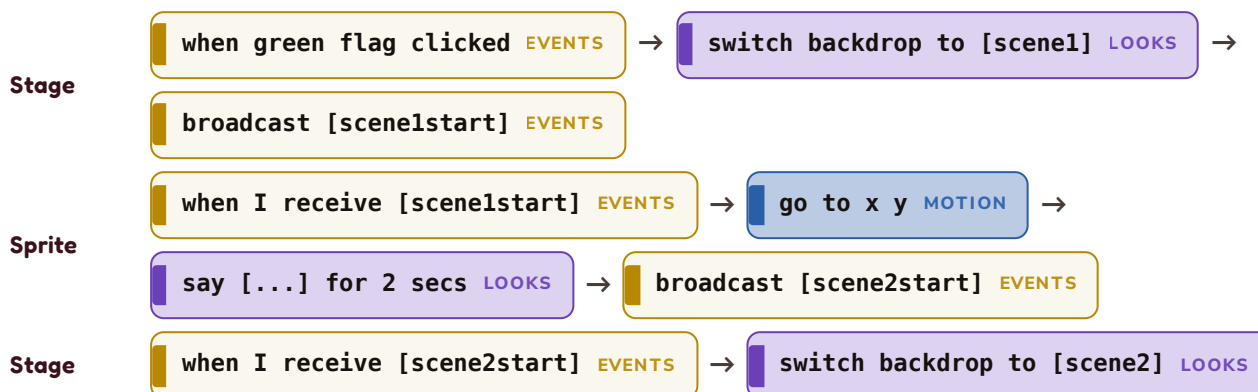
A wand sprite and two or three victim sprites.



Add a second message, `undo`, on a different key, that puts everything back. Making both directions work is what forces them to think about state.

Activity B – Two-Scene Story (16 min)

Two backdrops, two sprites.



A short story that runs itself. It's the first time they build something where the *sequence across sprites* is the interesting part.

0:55–1:05 – Break

1:05–1:25 – Project build: lives, game over, and a start screen

This is a bigger build than usual. If time is short, do lives and game over, and push the title screen to homework.

Step 1 – Lives

1. Make a variable `lives`. `set lives to 3` under the green flag.
2. In the bad object's catch `if`, add `change lives by -1`.
3. On the player (or stage):



Step 2 – Game over

1. Add a "Game Over" backdrop (paint one with text, or use a plain one and a text sprite).
2. On the stage: `when I receive [gameover]` → `switch backdrop to [Game Over]` → `stop all`.
3. On each falling object: `when I receive [gameover]` → `hide`.

Note the ordering trap: `stop all` stops *everything*, including the scripts that were about to hide the sprites. Either hide first and stop last, or use a `wait 0.1` before the `stop all`. This is a lovely real bug and worth letting them find.

Step 3 – Title screen (if time)

1. A "Title" backdrop with the game's name.
2. `when green flag clicked` → `switch backdrop to [Title]` → `say [Press S to start]`.
3. `when [s] key pressed` → `broadcast [startgame]`.
4. Move everything that was under `when green flag clicked` to `when I receive [startgame]`.

That last move is conceptually significant and they'll feel it: the green flag no longer starts the game, it starts the *title screen*. The game has states now.

Save.

1:25–1:30 – Wrap

Save and back up. Today's project has more scripts spread across more sprites than any previous week, and a lost file now costs real work.

Homework

1. **A way to win.** Pick a target score and broadcast a `youwin` message when the player reaches it, with its own backdrop.
2. **Restart.** Add a way to play again without clicking the green flag — a key press or a clickable button that resets the variables and broadcasts `startgame`.

Bonus challenges

1. Add a "3 — 2 — 1 — GO!" countdown between the title screen and the game starting.
2. Give each backdrop its own music, starting and stopping on the right broadcasts.

3. Add a boss object that appears only when the score passes 20.
4. Try `broadcast and wait` instead of `broadcast` somewhere and work out what's different. (Hint: it doesn't move on until every receiver has finished.)

Common bugs this week

Symptom	Cause
Broadcast does nothing	Message name typo, or two near-identical messages in the dropdown
Sprites stay visible after game over	<code>stop all</code> ran before the <code>hide</code> scripts got a chance. Add a short wait, or hide before stopping
Sprites are invisible when the game restarts	They were hidden at game over and never <code>show</code> n again. Add <code>show</code> under the start
Lives goes to -1, -2, -3	The <code>if lives = 0</code> check missed the exact moment. Use <code>< 1</code> instead of <code>= 0</code>
Game over fires immediately	<code>lives</code> wasn't set to 3 before the check started running
Everything stopped and won't restart	<code>stop all</code> really does stop everything, including restart scripts. Restart needs to be on a <code>when key pressed</code> hat, which still fires

The `= 0` versus `< 1` one is genuinely instructive. Exact equality checks are fragile because they only catch one specific instant. It's a real lesson in defensive programming, delivered at exactly the right moment.

Notes for the second volunteer

Today's debugging is different in kind from previous weeks: the code is *correct* in each sprite but the *interaction* between sprites is wrong. Teach them to trace the message — "who shouts this, and who's listening?" — rather than staring at one script. Getting a child to draw the messages on paper with arrows works remarkably well.

Links

- [Broadcasting spells](#) — Raspberry Pi Foundation, almost exactly the Magic Wand activity with more depth
- Skeleton: `Sky Catcher – end of Week 6` in the club studio